

Table of Contents

Spike reports..... 1

 Planned 1

 Conventions..... 1

Spike reports

Time-boxed investigations run to resolve a specific uncertainty before it can affect design or implementation. Each produces a report with a stated method, raw output, and limitations.

ID	Spike	Date	Status	Outcome
NOVA-SPK-001	Local model selection under a 6 GB VRAM ceiling	2026-09-07, completed 2026-09-08	Complete	gemma3:4b. 100 of 100 schema-conformant across five candidates, 17.3 s median, 2.9 GB fully GPU-resident. The 8B class spills a third onto the CPU for no quality gain
NOVA-SPK-002	Embedding model and vector dimension	2026-09-08	Complete	embeddinggemma at 768 dimensions. Leads every discriminating retrieval metric across five candidates, 77 ms warm, co-resident with the generation model inside 6 GB
NOVA-SPK-003	Container-to-host inference reachability and datastore provisioning	2026-09-09	Complete	Both reachable without configuration. Docker Desktop's host proxy reaches a loopback-bound Ollama, so the anticipated binding change is unnecessary. vector(768) indexes under HNSW and answers correctly

Planned

Spike	Resolves	Blocking
Hybrid lexical and vector ranking	NOVA-SPK-002 follow-up 3	No. Deferred until authentic corrections exist

Conventions

- A spike is time-boxed before it starts. Exceeding the box is itself a finding
- Every report states its method, its sample size, and its limitations. A measurement without a stated sample size is not a result

- Harnesses and raw output are committed under artifacts/ so that any figure can be reproduced
- A partial spike is reported as partial. Unmeasured candidates are listed as unmeasured rather than omitted
- Recommendations arising from a small sample are marked provisional and do not become decisions until confirmed